PROVEIT RESEARCH NOTE

Further Lean proof simplification

Controller arithmetic and subset coverage

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Main finding. Allowing one gap in a finite interval gives every fixed-cardinality subset sum in a cyclic group, for every cardinality $0 < a < T$. Coprimality is unnecessary. Applied to the existing affine power coordinates, this removes the optional Boolean correction from the arithmetic coverage theorem, provided its cardinality bound permits $a = C_0$.
Other findings. The fixed CRT components need periods, while the auxiliary prime-power component needs exact order. A single fixed residue modulo $94517 \cdot 120199$ proves the entire two-parameter family of controller coefficients is coprime to the required moduli. An all-exponent divisibility law reorganizes the prime-power order proof without a zero/successor split.
Review baseline. `9f35798db4a63c25e435fca14d75931144657d7e`. The initial audit covered 23 placed packets and five incoming archives. A final refresh through commit `f00ba53b331833c6cb64707bd90e48489e5610f5` reviewed the expanded 36-packet corpus and new controller clients. Earlier proposals are treated as available even when marked pending or deferred.

This report proposes proof changes and supplies reproducible Lean experiments. It does not assert new mathematical priority, a completed Diophantine consistency equivalence, or a measured whole-project speedup.

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1 Result, scope, and priorities

The most valuable simplification is to strengthen the finite subset theorem that the controller consumes. The existing proof realizes a sum by translating a consecutive interval. Translation changes its sum by a times the shift, so that particular construction needs a invertible modulo T. Changing the shape of the subset by just one gap removes that restriction. Consequently, the arithmetic coverage proof no longer needs a preliminary Boolean choice whose purpose is to make the cardinality coprime to an auxiliary prime power.

This is a concrete improvement over the stipulated prior-work baseline. It does not rename an old scanner, coefficient, witness-transport, or finite-table abstraction. The new proof applies to cardinalities that the old construction cannot handle. Sections 3 and 4 give the explicit witness and its exact interface consequence.

Priority	Proposal	What disappears from the relevant proof path
P1	Arbitrary proper cardinalities in cyclic subset sums	Cardinality coprimality and, with the endpoint change, the optional Boolean correction.
P2	Periods for fixed CRT components	Proofs of minimal multiplicative order where consumers only need an annihilating exponent.
P3	One residue for the complete controller coefficient	Separate dimension, quadratic-factor, coupling-factor, and coprimality arguments at each of three primes.
P4	Uniform annihilator law for the principal-unit tower	The $k = 0$ versus successor split in the exact-order proof.
Companion	Nonzero first lift as the public seed	Repeated natural-subtraction and divisibility bridges at the current-generator boundary.
Optional	Translating a specified seed	Interval-specific triangular arithmetic when a shape constraint makes translation useful.

P1–P4 are the primary directions. The companions target overlapping code and are not counted as additional independent source savings. In particular, the arbitrary-seed translation and unrestricted-cardinality constructions target the same interval machinery. Their deletion opportunities are not additive. A prime-support diagnostic suggestion was withdrawn after the final refresh found it in an earlier submitted packet.

The proof checks are pinned to the original revision; the final novelty and client review uses the refreshed revision. The Lean project pins Lean 4.32.0 and mathlib commit `81a5d257c8e410db227a6665ed08f64fea08e997` in `lean-toolchain` and `lake-manifest.json`. Validation receipts, source hashes, and the precise replay scope are supplied with the archive; Section 9 explains their interpretation.

2 Excluding the previous proposals

2.1 Final refresh and a withdrawn candidate

The final freshness check found 25 additional commits, through `f00ba53b3318`. The placed corpus now contains 36 packets. Five of the 13 new placements are the already-audited incoming archives; the other eight are new reports, `zc` through `zj`. The final exclusion review covers this delta as well. The exact original inventory below is retained as provenance, and the additional hashes and comparisons are supplied in the refresh audit. That delta contains 34 primary TeX components and 219 searched textual members. The five incoming relocations were checked byte-for-byte for their retained members; earlier omitted material remains part of the assumed baseline. See `evidence/FINAL_REFRESH_NOVELTY_AUDIT.txt` and `evidence/refresh_delta_inventory.json` for the final decisions.

The refresh identified a real overlap: packet `zd` already proposes finite prime-support certificates for multiplicative-order diagnostics. That candidate was therefore removed from this report's recommendations and from the delivered Lean experiments, despite its successful local check. All 15 source/configuration inputs used by the focused replay are unchanged between the two pins. The proof receipts remain anchored to the original pin; the novelty and client-impact review uses the refreshed corpus.

The newer `zh` controller proposal still keeps the Boolean adjustment and coprime-cardinality path, and its Hs simplification works on the separate prime residues. These are assumed available here. They do not supply P1's noncoprime subset theorem or P3's whole-coefficient residue transport.

2.2 What was audited

At the initial pin, the supplied exclusion locations contain 23 placed packets, named `2026-10-26a` through `2026-10-26w`, and five additional proof-simplification ZIPs in `docs/incoming`. The inventory records 74 primary TeX files or components, their paths, SHA-256 hashes, sizes, and section headings. Targeted searches cover 1,295 textual members, including supporting notes and archived source snapshots. That number describes the search scope; it is not a claim that every word of every duplicated snapshot was read. Report-section review and mechanism-level comparisons supplement the searches.

The five incoming files are:

- `Lean_Proof_Simplification_Post_Adoption_2026-10-26.zip`;
- `ProveIt_Additional_Lean_Simplifications_2026-10-26.zip`;
- `diophantine-lean-further-deltas-2026-10-26.zip`;
- `diophantine-lean-proof-refinements-2026-10-26.zip`;
- `proveit-lean-structural-simplifications-2026-10-26.zip`.

The machine-readable evidence is in `evidence/exclusion_inventory.json`, `evidence/exclusion_taxonomy.json`, and `evidence/prior_report_headings.tsv`. The large incoming README also links unrelated historical research; the audit does not claim to reread every transitively linked manuscript. The novelty claim is relative to the directly audited simplification corpus, not to all mathematical literature.

2.3 How "assume already applied" was interpreted

Every earlier proposed proof interface is treated as available, including optional, deferred, uncompiled, or unselected ideas. This is a conceptual union of prior methods. It is not a claim that all competing historical patches can be combined into one compiling implementation. Accordingly, this report does not subtract lines from the current tree and label the result an exact saving against an imaginary fully merged tree.

Excluded families include list and decoder APIs; scanner and history fusion; primitive-recursive congruence; assignment naturality; common DAG, gate, residual-square, and degree machinery; coefficient fibers and reflection; carry and cyclic-word interfaces; Pell sign, minimum, and perturbation arguments; internal-model arithmetic specifications; finite inverse invariance; Wang-tile fibers, helical periods, shared edges, pulse rows, receiver causality, and operating-row recognition. Earlier generic advice to use finite checks, weaken an unused premise, or introduce an equivalence is also not presented as a discovery here.

The retained proposals have more specific mechanisms: noncoprime fixed-cardinality cyclic subset sums; the exact separation between the fixed and auxiliary order obligations; erasure of both controller exponents in one residue ring; and a uniform all-exponent divisibility interface for LTE. The accompanying audit gives the nearest old ideas and the distinction for each proposal.

3 P1: all proper cardinalities realize every cyclic sum

3.1 The limitation of translated intervals

`ControllerIntervalSums.lean` constructs a subset by translating $\{0, 1, \dots, a-1\}$. Its sum is

$$a\,s + \sigma_a, \qquad \sigma_a = \sum_{i=0}^{a-1} i = \frac{a(a-1)}{2}.$$

Solving for the shift $s \in \mathbb{Z}/T\mathbb{Z}$ requires multiplication by a to be surjective. The existing theorem therefore uses $\gcd(a, T) = 1$ and an explicit unit inverse. The division by two is correctly performed in the naturals, not in the residue ring; that existing safeguard is not itself a new proposal.

The unit condition is a restriction on this family of translated subsets, not on the existence of a subset with the requested cardinality and sum.

Theorem 3.1 (Fixed-cardinality cyclic subset sums)**.** *For integers T, a with $0 < a < T$ and any $u \in \mathbb{Z}/T\mathbb{Z}$, there is a subset $S \subseteq \mathbb{Z}/T\mathbb{Z}$ such that*

$$|S| = a, \qquad \sum_{x \in S} x = u.$$

No coprimality, primality, or prime-power hypothesis is required.

3.2 An explicit witness with one gap

First prove a natural-number statement. For every $0 \le \delta \le a(T-a)$, the integer $\sigma_a + \delta$ is the sum of an a-element subset of $\{0, \dots, T-1\}$. Write

$$\delta = aq + r, \qquad 0 \le r < a,$$

and take

$$x_i = i + q + \mathbf{1}_{\{i \geq a-r\}}, \qquad 0 \leq i < a. \tag{1}$$

The sequence is strictly increasing: consecutive entries differ by one, except that a single jump may have size two. Its sum is

$$\sum_{i<a} x_i = \sigma_a + aq + r = \sigma_a + \delta.$$

The bound on δ implies $q \leq T - a$. If $q = T - a$, then $r = 0$; otherwise $q \leq T - a - 1$. In the first case the largest entry is $T - 1$; in the second it is at most $(a-1) + (T - a - 1) + 1 = T - 1$. Thus all entries are valid distinct representatives below T.

For the modular theorem, take δ to be the canonical representative of $u - \sigma_a$ modulo T. It is at most $T - 1$, and

$$a(T-a) - (T-1) = (a-1)(T-a-1) \geq 0. \tag{2}$$

The natural theorem therefore applies. Casting the chosen representatives to $\mathbb{Z}/T\mathbb{Z}$ preserves distinctness and gives the desired sum.

This proof is elementary and constructive. It uses a quotient and remainder, a finite image, an injectivity proof, and a suffix-cardinality sum. It needs no additive-combinatorics theorem and does not infer a field structure on `ZMod T`.

3.3 A small example that distinguishes the methods

For $T = 4$ and $a = 2$, translated consecutive pairs have sums $2s + 1$ and therefore reach only odd residues. Formula (1) reaches all four residues:

Target u	Selected representatives	Sum modulo 4
0	$\{1, 3\}$	0
1	$\{0, 1\}$	1
2	$\{0, 2\}$	2
3	$\{1, 2\}$	3

This is the relevant gain: an even cardinality modulo four becomes usable. The method does not just shorten an inverse calculation that remains necessary elsewhere in the same argument.

3.4 Lean decomposition and exact targets

`lean/RestrictedSubsetSums.lean` separates the proof into `staircase_strictMono`, `staircase_sum`, `staircase_lt`, `exists_nat_subset_sum`, `window_covers_modulus`, and `exists_subset_card_sum`. The natural theorem permits $a = T$ with offset zero. The modular theorem correctly requires $a < T$: a subset of size T is the whole group and has only one possible sum.

The first replacement target is the solver behind `ControllerIntervalSums.targetInterval`. In `ControllerOrbitTransport.lean`, the affine-sum transport remains useful; its subset input changes from the inverse-selected interval to the new staircase subset. The source search finds that the direct non-check client of `targetInterval` is this orbit-transport module. Historical explicit interval APIs can be retained as compatibility results if they are still wanted, but retaining every one reduces the achievable source deletion.

4 The affine consequence: remove the Boolean arithmetic correction

4.1 The new endpoint

`ControllerResidueLift.lean` currently proves `exists_subset_boolean_sum`. Its hypotheses include a prime p, $p \mid C_0$, $T = p^n$, an auxiliary weight with nonzero projection modulo p, and an injective coordinate map. Its conclusion uses a Boolean coefficient $e \le 1$ to arrange a cardinality $0 < a < C_0$ coprime to T.

With the previous section's theorem, the arithmetic endpoint can instead be:

Theorem 4.1 (Boolean-free affine coverage)**.** *Suppose $0 < C_0 < T$, $\iota : \mathbb{Z}/T\mathbb{Z} \to A$ is injective, and weights satisfy*

$$w(\iota(z)) = 1 + C_0 z \quad \text{in } \mathbb{Z}/(C_0 T)\mathbb{Z}.$$

Then every $t \in \mathbb{Z}/(C_0 T)\mathbb{Z}$ is the weight sum of an actual finite subset $S \subseteq A$ with

$$1 \le |S| \le C_0.$$

The expression $C_0 z$ denotes the well-defined additive map from $\mathbb{Z}/T\mathbb{Z}$ to $\mathbb{Z}/(C_0 T)\mathbb{Z}$; it does not require choosing a ring homomorphism between these residue rings in the wrong direction.

Proof. Let a be the least positive representative of the projection of t to $\mathbb{Z}/C_0\mathbb{Z}$. Thus $1 \le a \le C_0$ and $t - a$ has zero projection. There is a coordinate $u \in \mathbb{Z}/T\mathbb{Z}$ with $t = a + C_0 u$. Since $a \le C_0 < T$, the cyclic subset theorem supplies a subset $U \subseteq \mathbb{Z}/T\mathbb{Z}$ of size a and sum u. Set $S = \iota(U)$. Injectivity preserves cardinality, and additivity gives

$$\sum_{x \in S} w(x) = \sum_{z \in U} (1 + C_0 z) = a + C_0 \sum_{z \in U} z = a + C_0 u = t.$$

□

4.2 Why the non-strict endpoint is necessary

The existing output bound $a < C_0$ cannot be preserved while removing the correction term for all targets. Every weight projects to 1 modulo C_0, so a subset sum projects to its cardinality. A target with zero projection requires $a \equiv 0 \pmod{C_0}$. Its least positive permitted cardinality is therefore C_0 itself.

The checked adapter in `lean/AffineSubsetCoverage.lean` reuses the existing `exists_coordinate` and `scaledCoordinate`. The old coordinate lift uses a representative $a_0 < C_0$. When $a_0 = 0$, the adapter sets $a = C_0$ and replaces the coordinate u by $u - 1$; additivity shows $C_0 + C_0(u - 1) = C_0 u$. This explicitly handles the endpoint rather than asking simplification tactics to hide a change in the theorem statement.

If a downstream capacity theorem truly requires $a < C_0$, keep a Boolean choice in that API. The new subset-sum theorem still removes the need to make a coprime to T, so that weaker migration can eliminate the prime-power cardinality-selection argument without changing the output bound. The fully Boolean-free endpoint should be adopted only where the capacity contract accepts $a \le C_0$.

4.3 The actual controller parameters and remaining composition

The existing `CurrentPowerCoordinates2011.lean` supplies the relevant orbit with

$$p = 120199, \quad d = 94517, \quad C_0 = dp, \quad T = p^k.$$

The value of C_0 is dp, not d. Since the repository proves $p > d$, the inequality $C_0 < T$ holds at $k \geq 2$. The power-coordinate result also has smaller-exponent cases, but that does not make this subset theorem applicable when $T \leq C_0$. The orbit modulus here is $C_0 T = dp^{k+1}$. When identifying it with the coefficient modulus dp^n in P3, set $n = k + 1$; the coverage range is then $n \geq 3$. P3's separate unit theorem remains valid for every n, including zero.

For this arithmetic coverage result, the hypotheses $p \mid C_0$, primality, the prime-power presentation of T, and a separately supplied nonzero Boolean weight disappear. They remain where independently needed to prove the actual orbit or other controller facts. The new theorem does not show that every selected coordinate corresponds to a safe physical dummy position, nor that a specific shape constraint is preserved. It also does not by itself reduce the variables in the printed Diophantine equation.

The bound $C_0 < T$ has content. With $C_0 = T = 2$, the affine weights are $\{1, 3\}$ modulo four, whose subset sums miss 2. Thus simply removing that hypothesis would make the proposed statement false.

4.4 Compatibility with the newer physical-position proofs

The refreshed source now contains `CurrentControllerSubsets2011.lean`. Its `exists_normalized_subset` still calls the Boolean-based residue-lift theorem, so it is a direct new client for the proposed endpoint. The subsequent normalization by the unit B^2 is retained; its separate εB^3 summand disappears when $\varepsilon = 0$.

There is a useful compatibility fact in `CurrentControllerPositions2011.lean`: `support_safe` and `cyclicSupport_safe` quantify over an arbitrary finite subset of orbit indices. They require the geometric condition $r_w < k$, but no interval shape, coprimality, or strict cardinality cap. Their statements therefore apply to the staircase subset with the optional parameter set to zero. This is a source-level compatibility deduction from the maintained signatures, not a claim that the complete new composition has been rebuilt.

The worst-case physical support size does not increase. The existing `support_card` theorem gives $a + \varepsilon$. Under the old bounds, $a < C_0$ and $\varepsilon \leq 1$ imply $a + \varepsilon \leq C_0$; under the new bounds, $a \leq C_0$ and $\varepsilon = 0$ give the same cap. Thus allowing the arithmetic endpoint $a = C_0$ does not by itself demand more physical updates than the old interface already allowed.

The refresh also includes safe single-dummy increments, nonwrapping transport, and rank decrements in `CurrentDummySafety2011.lean` and `CurrentDummyMutation2011.lean`. These are existing ingredients to reuse, not missing local bounds to prove again. The unchecked migration step for this proposal is the finite-update composition: preserve the unused-dummy premise at each selected position, aggregate the increments, and bind that sum to the Boolean-free power-subset endpoint. The late-head scope addendum records the exact declarations and source identities.

5 P2: use periods at fixed factors, exact order at the auxiliary factor

5.1 The current over-specification

`CurrentControllerData2011.lean` proves the exact orders of 2 and $B = 2^{94517}$ modulo 47 and 2011. The production consumers of `B_order_47` and `B_order_2011` in the reviewed subtree are the fixed-prime power reductions in `CurrentControllerUnits2011.lean` and the projection lemmas in `CurrentPowerCoordinates2011.lean`. Those consumers only need

$$B^{23} = 1 \pmod{47}, \qquad B^{402} = 1 \pmod{2011}.$$

They never use the minimality of 23 or 402.

Two short monoid facts supply the appropriate interface:

$$x^m = 1 \wedge m \mid e \implies x^e = 1, \qquad x^m = 1 \implies (x^s)^m = 1.$$

The proofs are exponent laws; no group or commutativity assumption is needed. The file `lean/FixedPeriods.lean` proves both and the actual fixed-prime certificates. It also contains an alternative direct bounded modular certificate for the complete projection modulo $94517 \cdot 120199$. That calculation works in the finite residue ring and does not construct the unreduced natural number $2^{11360754366}$.

5.2 Where exactness must remain

`ControllerPowerCoordinates.lean` proves injectivity of the exponent parameter using the exact equation $\mathrm{ord}(g) = T$ and exponent representatives below T. Here an annihilating exponent is insufficient. For example, $1^3 = 1$ modulo nine, but the order is one and the three powers do not define an injection.

The correct division of responsibilities is therefore specific:

- fixed CRT components: prove the generator projects to 1;
- auxiliary prime-power component: prove exact order p^k;
- CRT composition: reuse the existing least-common-multiple order theorem;
- coordinate enumeration: consume the resulting exact full order.

The CRT least-common-multiple theorem is already present as `CurrentPowerCoordinates2011.order_product`. Recommending it again would not satisfy the user's request. Likewise, the report does not propose replacing the auxiliary exact-order premise by a mere period.

There are two distinct implementation goals. Rewriting the consumers removes their logical dependence on the fixed-prime minimality proofs. To remove the corresponding build dependency as well, move optional exact-order diagnostics into a separate module. If they remain in the same imported source file, Lean still elaborates them when that file builds. No timing claim is inferred from a shorter theorem dependency chain.

6 P3: one fixed residue proves the whole controller unit family

6.1 Erase both exponents in one residue ring

Set

$$p = 120199, \quad d = 94517, \quad M = dp = 11360848883, \quad B = 2^{94517}, \quad r = 2^{45919}.$$

The last equality matches the maintained factor radix^{977} because radix $= 2^{47}$ and $47 \cdot 977 = 45919$. Keep K as the original maintained kernel.

The coefficient in `CurrentControllerUnits2011.lean` is

$$\Gamma(r_w, n) = r(q^2 - 1)\big(1 + q(K + B^{p^{r_w}})\big), \qquad q = B^{p^n}.$$

The width exponent r_w and the cell-count exponent n are independent; neither is assumed positive.

In any monoid, a single fixed-point equation $x^p = x$ implies

$$x^{p^n} = x \quad \text{for every } n \in \mathbb{N}.$$

Induct on n and use $(x^{p^n})^p = x^{p^{n+1}}$. No primality, unit, or finite-field hypothesis is needed for this induction. For the actual parameters, bounded modular exponentiation verifies

$$(B : \mathbb{Z}/M\mathbb{Z})^p = B.$$

Both B^{p^n} and $B^{p^{r_w}}$ therefore reduce to B in the same ring, and the whole expression reduces at once:

$$\Gamma(r_w, n) \equiv \Gamma_0 := r(B^2 - 1)(1 + BK + B^2) \pmod{M}. \tag{3}$$

The natural subtraction is accounted for before casting to the residue ring: $B \geq 1$ implies $B^2 \geq 1$ and $(B^{p^n})^2 \geq 1$. Rewriting a natural subtraction as ring subtraction without those guards would be invalid. For example, the natural expression $0^2 - 1$ is zero, whereas the ring expression is -1.

6.2 One coprimality transport lemma

Lemma 6.1. *If* $a \equiv b \pmod{dp}$ *and* $\gcd(b, dp) = 1$*, then*

$$\gcd(a, dp^n) = 1 \quad \textit{for every } n \in \mathbb{N}.$$

Proof. Congruence preserves the gcd with dp, so a is coprime to both d and p. It is then coprime to p^n and to dp^n. □

This implication does not require d and p to be coprime or p to be prime. In the actual application, M is the radical of dp^n for $n > 0$. At $n = 0$ the target modulus has smaller prime support, and the same one-way implication still works. An unconditional converse would be false: 3 is coprime to $2 \cdot 3^0$, but not to $2 \cdot 3$.

Combining the lemma with (3) reduces the entire two-parameter unit theorem to one finite certificate for Γ_0.

6.3 The actual finite certificate

The current kernel residue facts in `CurrentKernelResidues2011.lean` are $K \bmod 47 = 26$, $K \bmod 2011 = 1995$, and $K \bmod 120199 = 114787$. CRT and finite arithmetic give:

Quantity modulo $M = 11360848883$	Representative
K	1491904576
B	10287028452
r	9930929229
Γ_0	706142772

Finally,

$$\gcd(706142772, 11360848883) = 1.$$

`lean/ResidueTower.lean` checks these computations in Lean while keeping K symbolic and using exactly the three existing residue facts as premises. It does not replace K by the smaller CRT representative as a natural number; only its residue is replaced inside modular arithmetic.

The generic theorem `gamma_coprime` and the actual-parameter adapter `current_gamma_coprime` include both zero-exponent endpoints. The adapter is a checked standalone result conditional on the existing kernel facts. A production integration must still apply it to the actual definition of K, identify the definitions of r and B, and rebuild the callers.

6.4 Expected proof simplification

The replacement targets four families, each with three prime-specific instances: `power_dimension_*`, `quadratic_mod_*`, `coupling_mod_*`, and `coprime_Gamma_*` lemmas. Their separate prime-specific derivations become one fixed-point proof, one whole-expression congruence, and one gcd certificate. Existing diagnostic residue statements can be recovered as projections if their public names should remain.

This proposal overlaps P2 in the fixed-prime dimension calculations; those lines must not be counted twice. The gain established here is the simpler dependency structure and the checked replacement core, not an already measured global deletion or build-speed improvement. The theorem proves a coefficient is a unit. It supplies no physical placement, safe increment, or final arbitrary-target rank correction.

7 P4: make the all-exponent annihilator law the LTE interface

7.1 A uniform theorem

For an odd prime p and a natural g with

$$p \mid g - 1, \qquad p^2 \nmid g - 1,$$

the maintained `PrimePowerOrder.lean` proves exact lifting at exponents p^k, then derives exact order by splitting k into zero and successor. The proposed interface exposes the whole divisibility predicate first:

$$p^{k+1} \mid g^m - 1 \quad \Longleftrightarrow \quad p^k \mid m, \qquad k, m \in \mathbb{N}. \tag{4}$$

For positive m, LTE says

$$v_p(g^m - 1) = v_p(g - 1) + v_p(m) = 1 + v_p(m),$$

and comparing valuations gives (4). The formal proof uses extended multiplicity so that $m = 0$, where both divisibilities hold, is included uniformly. The natural-subtraction guard still comes from $g > 1$, which follows from $p^2 \nmid g - 1$.

Let $o = \mathrm{ord}(g \bmod p^{k+1})$. Applying the forward direction to exponent o gives $p^k \mid o$, and applying the reverse direction to exponent p^k gives $o \mid p^k$. Antisymmetry proves $o = p^k$. This avoids the specialized prime-power order criterion and the $k = 0$ case split.

This is not a stronger final mathematical endpoint than the existing exact-order theorem: the latter already implies (4) through the usual order-divides-exponent characterization. The new contribution is the organization of the proof from LTE, giving a reusable all-exponent interface for essentially the same source size. The supplied refactor preserves the original public statements and names; exact helper-inclusive size and validation are recorded with the patch.

7.2 Accept a nonzero first lift directly

The current auxiliary generator is $g = B^{120199-1}$, and its checked remainder is

$$g \bmod 120199^2 = 7300767062 = 1 + 120199 \cdot 60739.$$

`lean/LiftSeed.lean` gives a small generic adapter:

$$1 < p, \quad c > 0, \quad g \bmod p^2 = 1 + pc \quad \Longrightarrow \quad g > 1,\ p \mid g - 1,\ p^2 \nmid g - 1.$$

The residue lower bound yields $g > 1$. Reducing again modulo p gives the first divisibility. The second would force the remainder modulo p^2 to be one, contradicting $pc > 0$. A separate $c < p$ premise is unnecessary: the fact that $1 + pc$ is a remainder already supplies that bound.

The target is `CurrentAuxiliaryOrder2011.lean`, which currently proves the positivity and divisibility bridges separately. This adapter aligns the theorem input with the finite certificate already available. The generic helper has a real source cost; at its first use alone, a large net line reduction is not claimed.

Oddness and primality remain necessary for the order corollary. Taking $p = 2, g = 3$ gives the same nonzero first-lift pattern, but the order modulo eight is two instead of four. Taking $p = 9, g = 28$ likewise shows that an odd composite base cannot replace the prime assumption.

8 A companion interface when a specified subset shape matters

P1 permits the subset shape to change. If an application instead needs a particular seed S, translation gives a separate useful interface:

$$|(|t + S) = |S|, \qquad \sum_{x \in t+S} x = |S|\, t + \sum_{x \in S} x.$$

When multiplication by $|S|$ is surjective, solve for t to prescribe the sum. Any translation-invariant property of S is preserved. `lean/SubsetTranslation.lean` proves this for arbitrary additive commutative groups, then specializes to $\mathbb{Z}/T\mathbb{Z}$ using coprimality.

This companion avoids the triangular formula and allows a noninterval seed, but it does not remove coprimality for translates of that one seed. The $T = 4$, $S = \{0, 1\}$ example explains exactly why. A physical-safety property is not automatically translation-invariant; that premise must be proved for the intended placement model. This companion and P1 target overlapping interval machinery, so their savings should be counted once.

9 Validation and reproducibility

9.1 What was actually checked

The seven delivered proof experiments use Lean 4.32.0 and the repository-pinned mathlib revision. They are ordinary kernel-checked Lean proofs. The final receipt inventory distinguishes standalone generic theorems, a checked adapter using freshly compiled repository prerequisites, and a focused production module replacement.

Artifact	Checking scope
`RestrictedSubsetSums.lean`	Passed; explicit natural construction and all proper positive cyclic cardinalities, including noncoprime examples.
`AffineSubsetCoverage.lean`	Passed against four freshly compiled unchanged controller modules and the checked cyclic subset theorem.
`FixedPeriods.lean`	Passed; generic period transport and actual modular certificates, with an exact-order counterexample.
`ResidueTower.lean`	Passed; whole-expression transport and actual numeric specialization conditional on the three existing kernel-residue facts.
`PrimePowerAnnihilator.lean`	Passed; all-exponent divisibility, modular powers, and uniform exact order.
`LiftSeed.lean`	Passed against a freshly compiled baseline principal-unit module.
`SubsetTranslation.lean`	Passed; additive-group theorem, invariant-property variant, and cyclic specialization.

The unchanged prerequisites `ControllerIntervalSums`, `ControllerOrbitTransport`, `ControllerAuxiliaryChoice`, and `ControllerResidueLift` were compiled freshly for the affine adapter. The uniform-order patch was checked separately under the real module name `DiophantineConsistency.PrimePowerOrder`; the unmodified `PrimePowerOrderChecks` module was then compiled against that fresh replacement. The patch also passes `git apply -check` at the pinned baseline. All six pre-existing public theorem signatures are preserved; the only additional public theorem is `pow_divisibility`.

The focused module changes from 80 to 79 physical lines, including the new helper. This is the one measured source-size comparison in the deliverable. Its principal benefit is a uniform theorem and fewer proof branches, not a large line deletion. Compiler wall times are retained as receipts, but the shared execution environment does not support a controlled timing comparison.

9.2 Trust and integration boundaries

The 27 printed axiom reports from the delivered prototypes and the six reports from the unchanged focused checks contain only the standard axioms `propext`, `Classical.choice`, and `Quot.sound`, or no axioms. The package's validation script rejects failed compiler runs and unexpected axiom dependencies. No supplied proof uses `sorry`, `admit`, a newly declared axiom, or `native_decide`.

The explicit distinction between checking levels matters. A compiled generic theorem with stated premises is not a rebuilt theorem about the entire production kernel. The residue adapter accepts precisely the already-proved kernel residue facts, while the affine adapter imports the existing coordinate infrastructure and changes the endpoint cardinality contract. The delivered patch has a narrower, genuinely replayed production scope. No full-project build or complete consistency-equivalence proof is claimed.

9.3 Replay

From the unpacked archive, use:

```
python3 scripts/replay.py /path/to/ProveIt
pdflatex -interaction=nonstopmode -halt-on-error report.tex
pdflatex -interaction=nonstopmode -halt-on-error report.tex
```

The replay script verifies the repository pin and needed source hashes, builds only the named prerequisites, compiles the proof experiments in an isolated output directory, then checks the replacement module and unchanged focused checks. It does not apply the patch or alter tracked source files. The scripts and receipts separate reproducibility from any future decision to adopt the changes.

The exclusion inventory can be regenerated without executing code from the earlier archives:

```
python3 scripts/build_exclusion_inventory.py \
  --repo /path/to/ProveIt --out regenerated-exclusion-audit
```

10 Suggested implementation sequence

1. Adopt the general cyclic subset-sum theorem and test its noncoprime clients. Keep the old interval constructors temporarily if callers need their explicit values.
2. Add the Boolean-free affine endpoint beside the old one. Make the $a \leq C_0$ capacity obligation visible at the exact physical composition site. Only then retire the old arithmetic Boolean detour where appropriate.
3. Replace fixed-prime exact-order dependencies by period or projection certificates. Split optional diagnostics into a separate module only if reducing build dependencies is part of the goal.
4. Introduce the whole-coefficient residue theorem and instantiate it with the three existing literal-kernel residue facts. Preserve r_w and n as independent parameters and retain their zero cases.
5. Apply the small uniform-annihilator patch and its focused checks. Choose the first-lift seed adapter when it simplifies the current-generator interface enough to justify the helper's maintenance cost.

For each adoption, measure source size after including helpers and retained compatibility declarations. Separately record build time, memory, and axiom dependencies. A smaller source file is not evidence of faster elaboration; an improved theorem interface can be worthwhile even when its first implementation has approximately the same line count.

11 Source and artifact guide

The claims about current code are grounded in the pinned source links throughout the report. The exclusion locations are the placed-packet index, the adoption record, and the incoming index. Standard library support is from the pinned multiplicative-order development, LTE and multiplicity development, and finite prime-factor interface.

The archive contains the complete LaTeX source and PDF, Lean experiments, the focused patch, proof-check receipts, a replay script, source and exclusion inventories, and independent mathematical-review notes. It is a reviewable research deliverable; none of these proposals was pushed to the repository.